Monte Carlo Simulation for Portfolio Analysis

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KEY FINDINGS

- Monte Carlo simulation (MC) is a one-size-fits-all approach for analyzing a broad range of pragmatic issues, notably in the field of portfolio management.
- Investors use MC for assessing the risk outlook of investment portfolios.
- MC serves as a validity check for analytical solutions to portfolio analysis and optimization problems and offers viable solutions to problems that are too complex to solve analytically.

ABSTRACT

Monte Carlo simulation is a powerful and pragmatic technique for assessing the performance outlook of investment portfolios. By imitating the price behavior of the assets in the portfolio via random draws from a probability distribution, one can generate fictive portfolio outcomes and, on that basis, derive return and risk forecasts. In this article, the authors lay out the basics of the analysis technique and provide a series of portfolio management examples.

Recognized as an effective analysis technique in principle since the late Renaissance,[1] Monte Carlo simulation (MC) reveals its power three centuries later in pace with the advancement of computer technology. Treating relatively complex and computation-intensive problems via MC in an ordinary Microsoft Excel spreadsheet on a personal computer has become feasible. The technique is thus easily accessible today, and the *modus operandi* is straightforward. In order to learn about a certain phenomenon, one can generate a large amount of possible outcomes for the phenomenon and make statistical inferences on those, typically starting by measuring the moments of the probability distribution, that is, the mean, the volatility, the skew, and the kurtosis. In this article, we expand on Simsek's (2023) introduction to Monte Carlo simulation for financial modeling and discuss how to use it for tackling common portfolio problems.

As a warming-up exercise, let us investigate the performance outlook of a basic investment. Assume we hold a single stock for one year that has a normal return distribution with a mean of 5% per annum and a standard deviation of 20%. We proceed by generating a large sample of possible investment outcomes for this stock, 100,000 draws typically, in an Excel spreadsheet, using the RAND function to generate random numbers between 0 and 1, followed by the NORM.INV function to convert those to

[1] In 1733, George-Louis Leclerc, the countess of Buffon, proposed a method for estimating pi (π) based on Monte Carlo simulation.

normally distributed variables. Without surprise, the sample mean will come close to 5% and its standard deviation close to 20%. Likewise, for the sample skew and kurtosis. The exercise becomes of interest when extending to alternative metrics and distribution functions, which we do next.

We look at two metrics that asset managers commonly use to capture the downside risk of investments, namely, the value at risk (VaR) and the conditional value at risk (CVaR) measures, at the 95% confidence level. They measure, respectively, the loss at the 5% probability point (α) and the expected loss beyond this point. It is easy to derive the MC estimates for these metrics by zooming in on the most adverse outcomes in the sample. The VaR should come close to its theoretical value of –28%, corresponding to the mean minus the Z-score times the standard deviation. The CVaR, specified in Equation 1 and estimated by taking the average over the 5% worst outcomes, comes close to –36%. Note that for the latter metric, it is simpler to obtain the MC estimate than it is to derive the theoretical value.

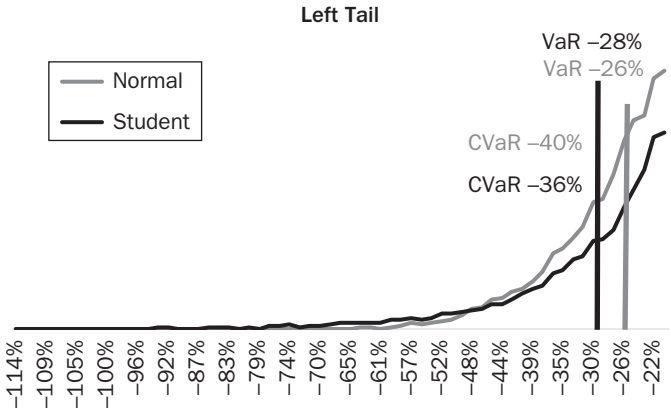
$$\text{CVaR} = \frac{1}{1-\alpha}\int_{-\infty}^{VaR} x \cdot p(x)dx \tag{1}$$

where $p(.)$ is the probability density function. The relative ease of the Monte Carlo simulation technique accentuates as soon as the normality hypothesis is relaxed. When assuming a Student *t*-distribution for instance, which according to many studies (Fama 1963) adapts better to the price behavior of stocks due to its higher kurtosis, the analytical formulas for the VaR measures become tedious to derive (Huschens and Kim 1999), whereas the procedure for obtaining MC estimates remains unchanged. It suffices to replace the Excel function NORM.INV by the function TINV for the *t*-distribution. Exhibit 1 gives simulation results based on draws from both distributions. It shows that, interestingly, the higher kurtosis of the *t*-distribution,

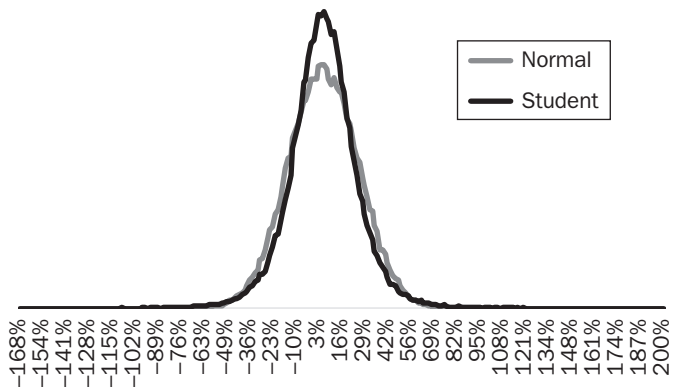
EXHIBIT 1

Stock Return Distribution and Value-at-Risk Measures

Panel A: Risk Measures and Left Tail



Panel B: Histograms



	Normal	Student
VaR	–28%	–26%
CVaR	–36%	–40%

NOTES: 100,000 draws from a normal distribution and from a Student t-distribution with 5 degrees of freedom, rescaled to align with the normal distribution in terms of mean and variance. Calculations made by the authors.

which is observable in Panel B, leads to a less severe VaR yet a more severe CVaR compared with the normal distribution (Panel A).

The same analysis setup suits for studying the impact of skewness on downside risk as well. We do this in the next section, where we look at the downside risk of investments in bonds that bear default risk. We then expand from single-asset investments to multivariate settings, whereby interdependencies among return variables come into play. Finally, we discuss how one can take advantage of the MC technique and take account of higher moments in portfolio optimization problems.

UNIVARIATE SIMULATION

In this section, we derive downside-risk estimates for bonds via Monte Carlo simulation. Mind that bonds bear two distinct risks, stemming from interest-rate movements and from credit events essentially, which are not evident to conciliate into a single metric. How to incorporate the possibility of default into the overall risk outlook of a bond investment? Taking time-series measures over historical returns would not do, as no default will have typically occurred over the observation period. In the spirit of Jobst, Mitra, and Zenios (2006), we proceed as follows: we simulate bond returns, convert the most adverse outcomes into default events and then take risk measures. To illustrate, we analyze two bonds, one issued by Apple Inc., which has a high credit rating, and another issued by GT Automotive, which has a much lower rating and thus presumably a sizeable left skew.

Besides the VaR and CVaR discussed earlier, we estimate the semi-volatility, denoted σ_{semi}, and the Sortino ratio (Sortino and van der Meer 1991), specified in Equations 2 and 3. The latter is similar to the Sharpe ratio except that the volatility parameter in the denominator is the semi-volatility.

$$Semi\text{–}volatility: \sqrt{\frac{1}{T}\sum_{r<mean}^{T}(r-mean)^2} \tag{2}$$

$$Sortino\ ratio: \frac{E[R]-R_{riskfree}}{\sigma_{semi}} \tag{3}$$

We generate a large quantity of return outcomes, 100,000 data points for each bond, by drawing from the normal distribution taking the historical mean and standard deviation as input parameters. We then incorporate default events into the samples in proportion to the respective default probabilities that credit rating agencies provide. We apply a default probability of 0.023% for Apple and of 0.942% for GT Automotive. Concretely, we replace the 23 worst outcomes for Apple by –70%, corresponding with a hypothetical total loss minus a recovery of 30%. For GT Automotive, we replace the 942 worst outcomes in the sample by –70%. Exhibit 2 displays the downside-risk measures calculated on the samples, with and without the simulated default events.

Exhibit 2 shows the impact of the left skew due to default risk on the downside measures. Estimating the metrics directly on historical returns data would produce figures close to those in the *without default* columns. Naturally, the figures are less severe than those in the *with default* columns that account for default and thus the left skew. For the Apple bond, the impact is limited: The CVaR estimate goes slightly up from a 5.5% loss to a 5.8% loss, and the semi-volatility goes up from 3.8% to 4.1%, while the Sortino ratio loses four basis points. Yet for the high-yield bond, the impact is substantial: The CVaR nearly doubles, the semi-volatility gains 3.5 percentage points, while the Sortino ratio loses 19 basis points. Note that the VaR does not alter per construction.

EXHIBIT 2

Downside-Risk Estimates Based on MC

	Apple Inc.		GT Automotive	
Metric	Without Default	With Default	Without Default	With Default
Mean	2.3%	2.3%	8.2%	7.8%
Standard deviation	3.8%	3.9%	11.0%	13.0%
VaR	–4.0%	–4.0%	–9.9%	–9.9%
CVaR	–5.5%	–5.8%	–14.5%	–23.6%
Semi-volatility	3.8%	4.1%	11.0%	14.5%
Sortino ratio	0.60	0.56	0.75	0.56
Sharpe ratio	0.60	0.58	0.75	0.63

NOTE: The nominator of the Sortino and Sharpe ratios is, for illustration purposes, simply the mean return, thus assuming zero risk-free rate.

SOURCES: Historical mean and standard deviation measured on monthly total return data provided by BoA Merrill Lynch between May 2013 and December 2019 for a bond issued by Apple Inc. and one by GT Automotive. Calculations made by the authors.

The interested reader may explore further and carry out tests in the same setting introducing both skew and kurtosis, the latter by drawing from leptokurtic distributions. There are, basically, three approaches for doing so. One can use the inverse cumulative distribution function, as is done in the introduction. If the desired distribution is not at hand, it may be possible to derive it from others that are available. In particular, there is the well-known Box–Muller (1958) transform that converts uniform draws into normally distributed variables. The *t*-distribution is the ratio of the normal- and the chi-squared distribution. Even the notoriously difficult fat-tailed and skewed Stable distribution is feasible to simulate, as it is a transformation of a uniform and an exponential distribution (Chambers et al. 1976).

The third method for making univariate draws is rejection sampling. In this approach, variates generated from an available distribution are either accepted or rejected based on a probabilistic test. The test is based on the ratio between the available and the desired probability density function and can be computationally quick. This route is especially useful when the normalizing constant of the desired function is difficult to derive. Marsaglia's polar method (Marsaglia and Bray 1964) is a well-known application of rejection sampling to simulate from a normal distribution. Rejection sampling can be applied to any distribution. The acceptance rate of candidate variates depends on how close the available and the desired distribution are, and therefore it is worthwhile, if one opts for this route, to spend time identifying a close distribution to the desired one.

Two remarks of a practical nature are in place. First, we remind researchers that, to ensure they can replicate any simulation they execute, the generated random numbers must be replicable. This is achieved in most computational software applications by setting a *seed* value from which the random numbers are generated; however, this functionality is not available in Excel. The RAND function refreshes the numbers every time the worksheet is calculated. In case Excel is used, we recommend, before starting the tests, copying the column of generated random numbers into a new column as values. It resolves the replicability imperative for academic research, and furthermore, it avoids the continuous refreshing, which slows down the calculus without benefit.

The second remark concerns the sample size of the simulation. With the exponential improvement in computer speed and power, it is usually computationally cheap to simulate a substantial number of observations. Having said that, a low-dimensional

test setting requires fewer observations than high-dimensional settings. There has been a trend of late among investment practitioners of using 10,000 observations when executing a Monte Carlo simulation. That may be sufficient for a univariate environment, but it is woefully short when working in multivariate space. Once the number of assets in the analysis exceeds five, the number of observations should be in the order of 100,000s or even 1,000,000s.

To help decide the sample size, several metrics are available that monitor output variability when simulating. Rather than using those, we find it more practical to run a simulation experiment several times and compare the summary statistics directly to see their variability. In our experience, running between 5 and 10 simulations is sufficient for such an assessment. While in the construction phase of a Monte Carlo model, one can increase the sample size until the variability of the output statistics falls within a predetermined tolerance level. Unless there is a substantial computation burden in producing the output, it is worthwhile keeping a copy of the simulated data from each run, which can serve in one mega simulation thereafter.

There are numerous texts detailing methods of univariate random number generation. As with many computational matters, Press et al.'s (2007) *Numerical Recipes* provides a good exposition of simulation and code snippets. For a more comprehensive and theoretical treatise, we find Ripley (1987) to be an exceptional book.

MULTIVARIATE SIMULATION

Turning next to multivariate settings, the interdependencies among the variables come into play, which makes the simulation more complex. The simplest multivariate case is drawing from the normal distribution, which involves the first two moments only, meaning that all dependencies are encapsulated in one correlation matrix. As soon as a distribution is governed by parameters involving skew and kurtosis, we need to go beyond that. For example, we might believe that two asset returns are correlated normally, but if we think that the return distributions exhibit high kurtosis, what to think of the likelihood of the assets experiencing extreme returns at the same time? Notwithstanding the normal correlation, one asset exhibiting an extreme return may not govern whether a second asset does. This might occur if the kurtosis stems from the occurrences of corporate actions such as buy-outs or rights issues. All of these dependencies need to be specified before the simulation can begin.

There are two main approaches for generating interdependent random numbers: variable transformation and using copulas. As to the first, in order to draw from the multivariate normal, one can transform a vector of independent normal variates by multiplying by the Cholesky decomposition of the covariance matrix to form a vector of dependent variates. Adepts of Excel can use the function MNORMRAND, which applies this method, that is available via the freeware add-in Real Statistics. To draw from a multivariate *t*-distribution, one can generate normal variates and scale them by simulated chi-squared variates (all doable in Excel). Even the multivariate stable distribution that exhibits both skew and kurtosis can be simulated by applying a linear transformation of independent stable variates (Press 1972).

A shortcoming of using fat-tailed distributions, such as the multivariate Student's *t* or the stable, is that it confines all asset return distributions to having the same fatness of tail, and when using Student's *t*, all must also be symmetric. A factor-model approach resolves this shortcoming. The asset returns, denoted by $n \times 1$ vector $\boldsymbol{r}$, are specified as

$$\boldsymbol{r} = \boldsymbol{\beta}^T \boldsymbol{X} + \boldsymbol{e} \tag{4}$$

where the $m \times 1$ vector $\boldsymbol{X}$ denotes the m factor returns, the $n \times m$ matrix $\boldsymbol{\beta}$ denotes the factor sensitivities of each asset return to each factor, and the $n \times 1$ vector $\boldsymbol{e}$ denotes the residuals, which are mutually independent and independent of the elements of $\boldsymbol{X}$. If the model has a single factor, $\boldsymbol{X}$ is a scalar thus independent of $\boldsymbol{e}$, and the variables can be simulated using univariate techniques enabling easy incorporation of nonnormality. If multiple factors are used that are uncorrelated, this method applies also. In this setting, $\boldsymbol{\beta}$ accounts for all dependency among the variables. In case of correlated factors, they can be simulated if they are assumed to follow a joint distribution that is readily available, such as the normal, t, or stable. Any distributional properties that are not present in the factor distribution can go in the elements of $\boldsymbol{e}$. The factor-model approach is a quick and easy way to incorporate nonnormality into a simulation of asset returns.

The variates can be further transformed by adding simulated shocks that occur with specified probabilities. These shocks can be events such as defaults, as in the earlier bond example, or corporate actions. We find that t-distributed data adjusted with an appropriately parameterized shock variable can fit return data reasonably well. An advantage of this method is that all manipulations are doable and traceable in Excel. A disadvantage is that the transformations introduce, potentially unintentional, distributional assumptions and miss subtler features evident in a dataset. Alternatively, normally distributed simulated variates can be transformed using a polynomial whose parameters are set such that the resultant variates exhibit the desired skew, kurtosis, and correlation matrix. See Vale and Maurelli (1983) and Headrick (2002) for details.

The second approach, using copulas, sees widespread use in finance. For further reading, we recommend Schmidt (2006), Aas (2004), Jouanin, Riboulet, and Roncalli (2003), and in particular the book by McNeil, Frey, and Embrechts (2005). We shall describe the copula method in detail before using it in a worked example.

Sklar's theorem (1959) says that every multivariate cumulative distribution function can be expressed by a function, termed a copula, taking as its arguments each of the univariate marginal cdfs. Hence, one can simulate multivariate variates as follows:

1. Draw from the multivariate copula, which can be a known distribution such as the Gaussian or Student t.
2. Transform the variates to probabilities using the inverse cdf.
3. Transform the probabilities into individual variates using each of the univariate marginal inverse cdfs.

This powerful method enables the generation of dependent variates from *any* marginal cdf, even nonparametric ones constructed from a historical dataset, and can therefore account for subtler elements of the marginal distributions. Copulas are very general and likely cover any context the financial researcher faces; indeed, the main practical challenge is to make the specification choices among the wide range of possibilities.

In step 3, we need the inverse of the marginal cdfs. Considering that taking the inverse can be computationally inexpensive, we suggest using the technique of kernel density estimation, whereby the cdf representing the sample dataset $x_1, \ldots, x_n$ is given by

$$F(x) = n^{-1} \sum_{i=1,n} K((x - x_i)/h) \tag{5}$$

where K is the cumulative density function of the kernel and h is the bandwidth parameter for which the standard "rule-of-thumb bandwidth estimator" is $h = 1.06\sigma n^{-1/5}$

where σ denotes the sample standard deviation. See Li and Racine (2007) for fuller details on kernel density estimation. Using Equation 5, the cdf, denoted $F(x)$, can be computed for any value x. A root-finding algorithm can be used to find the inverse cdf needed for the copula simulation.[2] We suggest computing the inverse cdf values for the probabilities 1%, 2%, …, 99%, which we denote $\xi_1, \xi_2, \ldots, \xi_{99}$. Then a random variate u, lying between 0 and 1, can be quickly transformed to be from the distribution with cumulative distribution function $F(.)$ as follows:

$$\begin{aligned} &\text{Compute } i = \lfloor 100u \rfloor \\ &\text{if } 1 \le i \le 98 \text{ then } x = \xi_i + z(\xi_{i+1} - \xi_i) \text{ where } z = 100u - i \\ &\text{if } i < 1 \text{ then } x = \xi_1 + a_L \ln(100u) \\ &\text{if } i > 98 \text{ then } x = \xi_{99} - a_H \ln(100(1-u)) \end{aligned} \tag{6}$$

where a_L and a_H are positive parameters set such that the marginal distribution exhibits the targeted skew and kurtosis.[3] The choice of a_L and a_H only affects 2% of the simulated variates, therefore this method is particularly useful if a simulated multivariate sample is required to exhibit a targeted skew and kurtosis, but the vast majority of the distribution follows the empirical kernel cdf, which may not exhibit the targeted skew and kurtosis.

In step 1, correlated copula variates are generated. Unfortunately these correlations will not be preserved in the final variates. Therefore, if the final variates are desired to exhibit specified correlations, then the required copula correlations to generate those specified correlations need to be derived. There are three common approaches to do this:

1. There is a large literature on estimating the copula correlations for a given data set; or
2. the correlations can be computationally computed, for example, using quadrature to solve the integrals of covariance; or
3. use Monte Carlo simulation and a root finder.

In the spirit of this article, and because we have the code for generating this Monte Carlo simulation (because that is our ultimate objective), the third approach can be convenient.

As for the univariate case, many open-source scripts exist to perform simulations, including using copulas. We stress in the strongest terms the need to test the output of a multivariate simulation prior to its use "in anger." At a minimum, the user should test the individual simulated asset returns against their marginal distributions and should test data dependencies across the assets by measuring the correlations and/or other suitable metrics of dependence, as the environment requires.

As to the simulation sample size, the question becomes critical when departing from the normal distribution. When using distributions of high kurtosis, by definition, they exhibit low likelihood but "black swan" events. Enough observations must be

[2] Such algorithms are available for many high-level programming languages such as Python or Matlab. To write one from scratch we refer the reader to *Numerical Recipes* by Press et al. (2007).

[3] For $x < \xi_1$ and $x > \xi_{99}$ we have approximated the cumulative distribution function by $F(x) = 0.01 \exp\{(x - \xi_1)/a_L\}$ and $F(x) = 1 - 0.01 \exp\{-(x - \xi_{99})/a_H\}$, respectively. Therefore, the a_H and a_L can be find using a basic root-finding algorithm using $E[X^k] = 0.01\ ((-1)^k \exp\{\xi_1/a_L\}\ a_L^k\ \Gamma(F_1^{-1}/a_L,\ k+1) + (k+1)^{-1}\sum_{i=1,98} (\xi_{i+1}^{k+1} - \xi_i^{k+1})/(\xi_{i+1} - \xi_i) + \exp\{\xi_{99}/a_H\}\ a_H^k\ \Gamma(F_{99}^{-1}/a_H,\ k+1))$ where $\Gamma(x,s)$ denotes the upper incomplete gamma function (see Abramowitz and Stegun 1972, Equations 6.1.5, 6.5.2, 6.5.3, and 6.5.13, for its computation).

simulated such that these events appear in the simulated data. Moreover, if there are multiple assets exhibiting heavy kurtosis, an even higher number of observations is necessary not only to ensure that these black swan events appear for a single asset but also to ensure that there are occurrences when these events occur simultaneously in multiple assets. We point the interested reader to the insurance literature where matters of dependency are central and therefore the topic of dependency specification is well developed.

In real investment contexts, forward-looking expected returns are often provided, for example via capital market assumptions (CMA). Even with the simple Gaussian assumption it is typical to use historical correlations and standard deviation but use forward-looking expected returns rather than historic ones. Sometimes the historical standard deviations are also replaced with forward-looking ones. We are at pains to point out that we are not restricted to using historical expected returns and standard deviations in our copula-based Monte Carlo simulation. Incorporating forward-looking expected returns and variances is simply done by transforming the variates generated from the empirical distribution as outlined earlier in this section (which uses historical expected returns and standard deviations) as follows: If we have a variate x from a distribution with historical mean μ_x and standard deviation σ_x, and we required a variate y from a distribution with forward-looking mean μ_y and standard deviation σ_y, then we compute

$$y = (\sigma_y / \sigma_x)(x - \mu_x) + \mu_y \tag{7}$$

A substantial benefit of such a transformation is that the original correlations, skew, and kurtosis are preserved. Thus, we have variates with forward-looking means and standard deviations but which exhibit all other distributional characteristics that appear in the historical data.

A MULTI-ASSET INVESTMENT EXAMPLE

To demonstrate the application of copula-based simulation, we provide a numerical worked example of a portfolio containing four asset classes: Treasuries (T), investment-grade corporate bonds (IG), high-yield corporate bonds (HY), and equity shares (EQ). We use US market indexes, namely the on-the-run five-year US Treasury bond index, the ICE BofA US IG bond index, the ICE BofA US HY bond index, and the S&P 500 equity index. We keep the asset universe small for illustration purposes but make note that the method applies to any type of asset and number of assets.

Exhibit 3 gives the descriptive statistics based on daily index returns from February 1, 1990, to July 7, 2024. Panel A shows large differences in skewness and tail behavior across the asset classes. Recalling that for a normal distribution, skew and excess kurtosis are zero valued, we see in the exhibit that the Treasury bond returns do exhibit a small amount of kurtosis. Abnormally negative Treasury returns occurred in many of the years in the dataset and were not associated with one or two specific events; conversely, abnormally positive returns only occurred in the Global Financial Crisis and during the Covid-19 period; indeed, all of the extreme positive Treasury returns relate to the epidemic.

Investment-grade bonds are somewhat left skewed and are more fat-tailed than Treasuries. The negative returns of this asset class are mainly associated with the global financial crisis and Covid, whereas the abnormal positive returns are all associated with Covid. High-yield bonds are particularly nonnormal, exhibiting a highly left

EXHIBIT 3

Descriptive Statistics of the Historical Data, Daily Returns from February 2, 1990, to July 1, 2024

	Treasury Bonds	Investment-Grade Bonds	High-Yield Bonds	Equity
Panel A: Skew and Kurtosis				
Skew	0.01	–0.56	–2.11	–0.18
Kurtosis	2.93	5.46	37.58	10.35
Panel B: Correlation				
Treasury bonds	1			
Investment-grade bonds	0.83	1		
High-yield bonds	0.04	0.32	1	
Equity	–0.19	–0.07	0.33	1

NOTES: Kurtosis shows excess kurtosis, in excess of 3. We calculated returns for the Treasury index on the bond yields, assuming that an auction takes place at the start of each month and thus that the coupon can be approximated by the start-of-month yield, then computed the dirty price during the month by discounting the cash flows at this yield. At the start of each month, the old bond is rolled into the newly issued five-year bond. We used price returns, not total returns that were unavailable to the authors at the time of writing, which serve the test purpose.

SOURCES: Federal Reserve's FRED database for the five-year US Treasury bond index and for the two corporate bond indexes; UK.investing.com for the S&P 500 Index price levels. Calculations made by the authors.

skew and substantial kurtosis. These characteristics are perhaps consistent with the notion that high-yield bonds behave like ordinary bonds until default becomes likely. We note with interest that the largest bond return occurred in Treasury bonds in 2009, after the Federal Reserve announced its intention to buy up long-term Treasuries. This highlights that the shocks causing abnormal price movements can occur in any market and need not only be driven solely by corporate news.

Equity returns have more kurtosis than IG bonds but substantially less than HY bonds. It is interesting to observe that despite the kurtosis, they are not particularly skewed; that is, an abnormal return has virtually equal chance of being positive or negative. Similar to other asset classes, the highest abnormal equity returns occurred during the Covid pandemic, with a significant number of high returns also observed during the financial crisis. Interestingly, these high returns mainly appeared in the first months of the crisis rather than during the recovery phase. There were also abnormally high returns in the recovery from the Asia crisis and the Russian crisis/Long-Term Capital Management (LTCM) blow up of 1997 and 1998, respectively. All the worst equity market returns occurred during Covid and the Global Financial Crisis, especially the latter.

In line with the stylized facts that we usually see in asset returns, the investment-grade and Treasury bond returns are highly correlated, and as the "safest assets" they are slightly negatively correlated with the riskier ones, while the high-yield returns are somewhat correlated to equity returns.

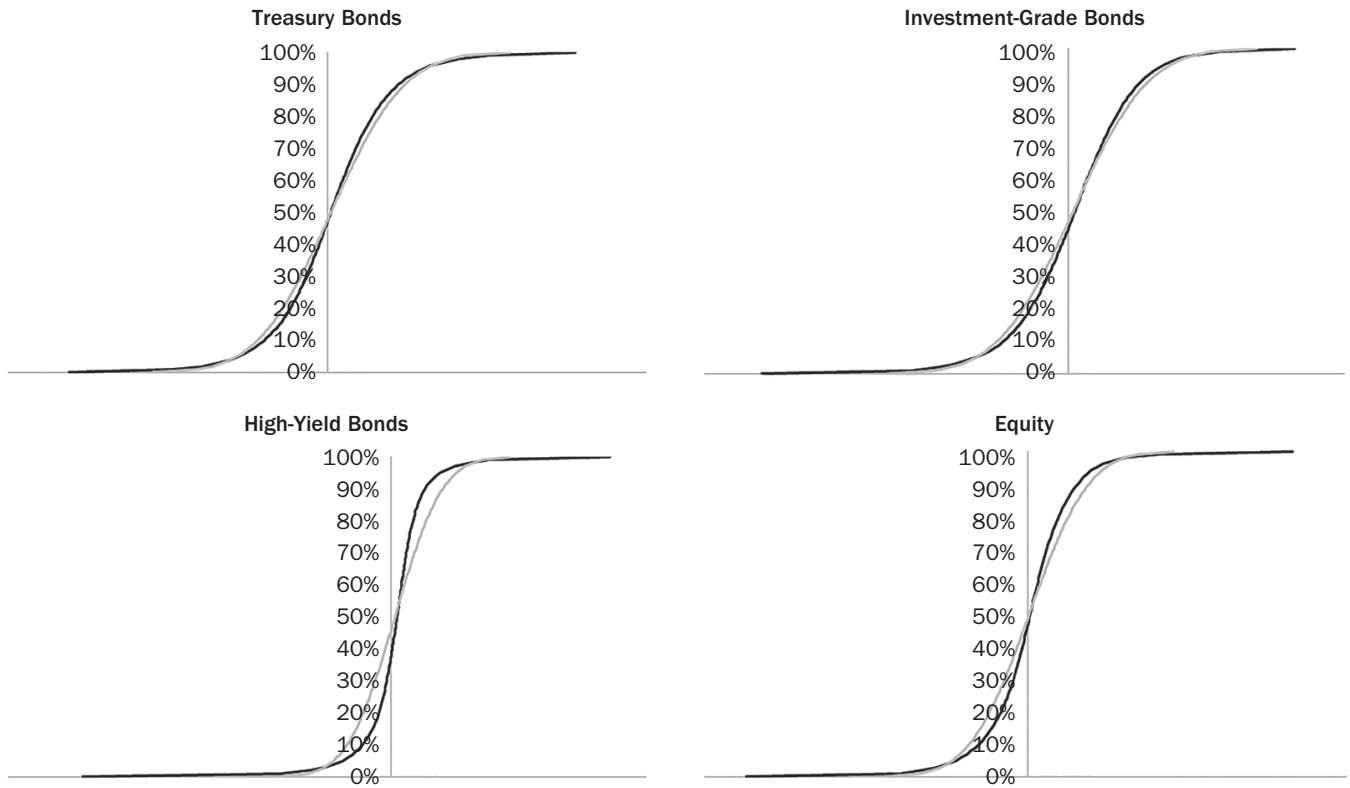
We demonstrate how to simulate returns from a four-dimensional multivariate distribution that represents the asset classes. These returns will exhibit the correlation matrix of Panel B in Exhibit 3 and the skew and kurtosis of Panel A and will have their entire marginal distributions equal to the empirically observed ones.

We computed the marginal cumulative distribution functions (cdfs) using the kernel estimation method, specifically Equation 5, with a Gaussian kernel, where $K(.)$ is the cdf of a standard normal distribution. (Note that because we have a large dataset, the choice of kernel is less critical.) The kernel-estimated empirical cdfs are shown in Exhibit 4, with the normal distribution cdf included for comparison.

We used a Gaussian copula. That is, in step 1, we generated four arrays of independent standard normally distributed data (using the Box–Muller transformation).

EXHIBIT 4

Cumulative Distribution Functions: Historical versus Theoretical Data



NOTES: See notes for Exhibit 3. The black lines indicate the historical distribution functions, and the gray lines the theoretical functions under the normality hypothesis. Calculations made by the authors.

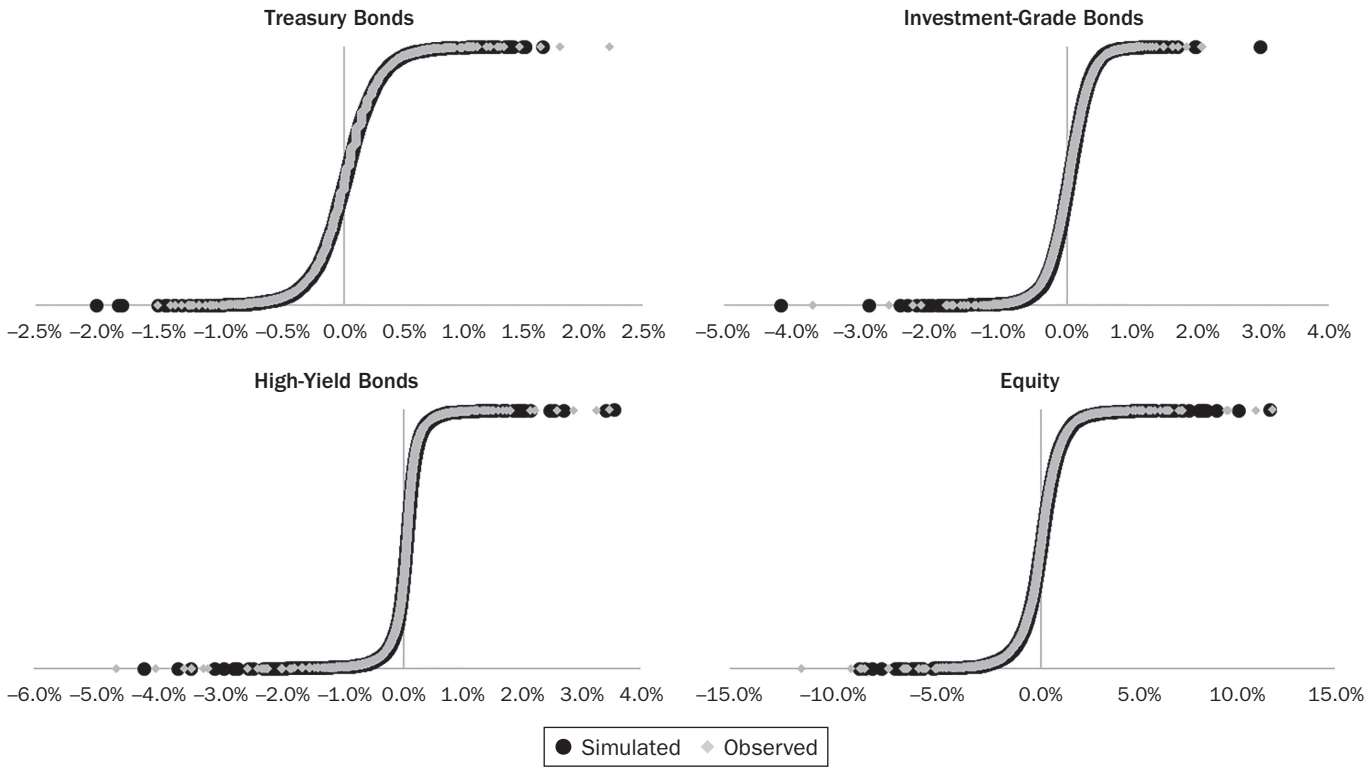
We multiplied the 4×1 vectors of independent variates by the Cholesky decomposition of the copula correlation matrix to form a vector of dependent Gaussian variates. These variates were transformed using the normal cdf function (approximated using Abramowitz and Stegun 1972, Equation 26.2.17). Finally, for each asset class, we transformed the cdf value into the required variate using Equation 6. To identify the copula correlations, on an asset pair basis, we simulated 10,000 asset returns and used a root-finder to find the copula correlation that yielded simulated final variates with correlations equal to those in Panel B of Exhibit 3.

We were now in a position to generate returns for the four asset classes. We simulated 250,000 returns for each using the copula approach described earlier (i.e., 1,000,000 variates in total). Exhibit 5 shows their cdfs alongside the observed data. A comparison between Exhibits 4 and 5 makes the benefit of the copula approach over the assumption of Gaussian returns immediately clear. Moreover, as tabulated in Panel A of Exhibit 6, these simulated returns exhibit skew and kurtosis consistent with the original data. A comparison with Exhibit 3 illustrates that the higher moments of the underlying data are preserved in the simulation; and, as tabulated in Panel B, the observed correlations are preserved in the simulated data as well.

In preparation for the next section where we carry out a portfolio optimization, we adopt forward-looking expected returns and risks. We take the annualized standard deviations of the return distribution of Treasury bonds, investment-grade bonds,

EXHIBIT 5

Cumulative Distribution Functions: Historical versus Simulated Data



NOTES: See notes for Exhibit 3. All historical data points have been used to produce the historical distribution function (in black), and 10,000 simulated data points have been used to produce the simulated function (in gray). Calculations made by the authors.

EXHIBIT 6

Descriptive Statistics of the Simulated Data

	Treasury Bonds	Investment-Grade Bonds	High-Yield Bonds	Equity
Panel A: Skew and Kurtosis				
Skew	0.00	–0.57	–2.15	–0.21
Kurtosis	2.92	5.51	38.31	10.80
Panel B: Correlation				
Treasury bonds	1			
Investment-grade bonds	0.83	1		
High-yield bonds	0.32	0.30	1	
Equity	–0.19	–0.07	0.32	1

NOTES: Kurtosis shows excess kurtosis, in excess of 3. Calculations made by the authors.

high-yield bonds, and equity to be 4.5%, 5%, 6%, and 20%, respectively. We then assume the market portfolio consists of 20% Treasuries, 20% investment grade, 10% high yield, and 50% equity. With these market weights, we assume the expected return to Treasuries to be 4% and use a Black and Litterman (1992) style model to derive expected returns for the remaining assets. The 1,000,000 variates were therefore scaled as per Equation 7.

PORTFOLIO CONSTRUCTION VIA MONTE CARLO SIMULATION

Statistical decision theory informs us that the best portfolio weights are those that maximize the expected utility of a portfolio. Unfortunately, given a vector of portfolio weights, we cannot generally derive the distribution of portfolio returns from the joint distribution of the individual assets returns. Only when the asset returns follow the class of stable distributions, which includes the normal distribution, is the portfolio distribution from the same distributional class as the individual assets and thus the portfolio return distribution can be derived. However, even if it were possible to derive this portfolio return distribution, for a given utility function, it is difficult to derive the distribution of the portfolio utility from the portfolio return distribution. Consequently, notwithstanding the theoretical validity of maximizing expected utility, in general it cannot be applied in practice.

Markowitz's (1952) modern portfolio theory (MPT) resolved the deadlock by offering a method of identifying best portfolio weights from a quadratic optimization. Because this optimization uses only the expectation and variance of portfolio return, it offers a useful method to construct portfolios and it still sees widespread use today. However, MPT is only consistent with decision theory when the asset returns obey to a joint normal distribution.

In his doctoral thesis, Buckle (1994) pointed out that if we have already simulated asset returns, we need not limit ourselves to MPT and can instead find portfolios of maximum expected utility. This is the case regardless of the distribution of the asset returns. The idea is that if we had a set of simulated asset return vectors $r_1, \ldots, r_T$, then given a weight vector, w, we can generate a set of portfolio returns, $R_1, \ldots, R_T$ where $R_i = w^T r_i$ (for $i = 1, \ldots, T$). Buckle (1994) noted that given this set of simulated portfolio returns, the expected utility of portfolio return can be estimated by Monte Carlo simulation, giving us

$$E[u(R)|w] \approx 1/T \sum_{i=1,T} u(R_i) = 1/T \sum_{i=1,T} u(w^T r_i). \tag{8}$$

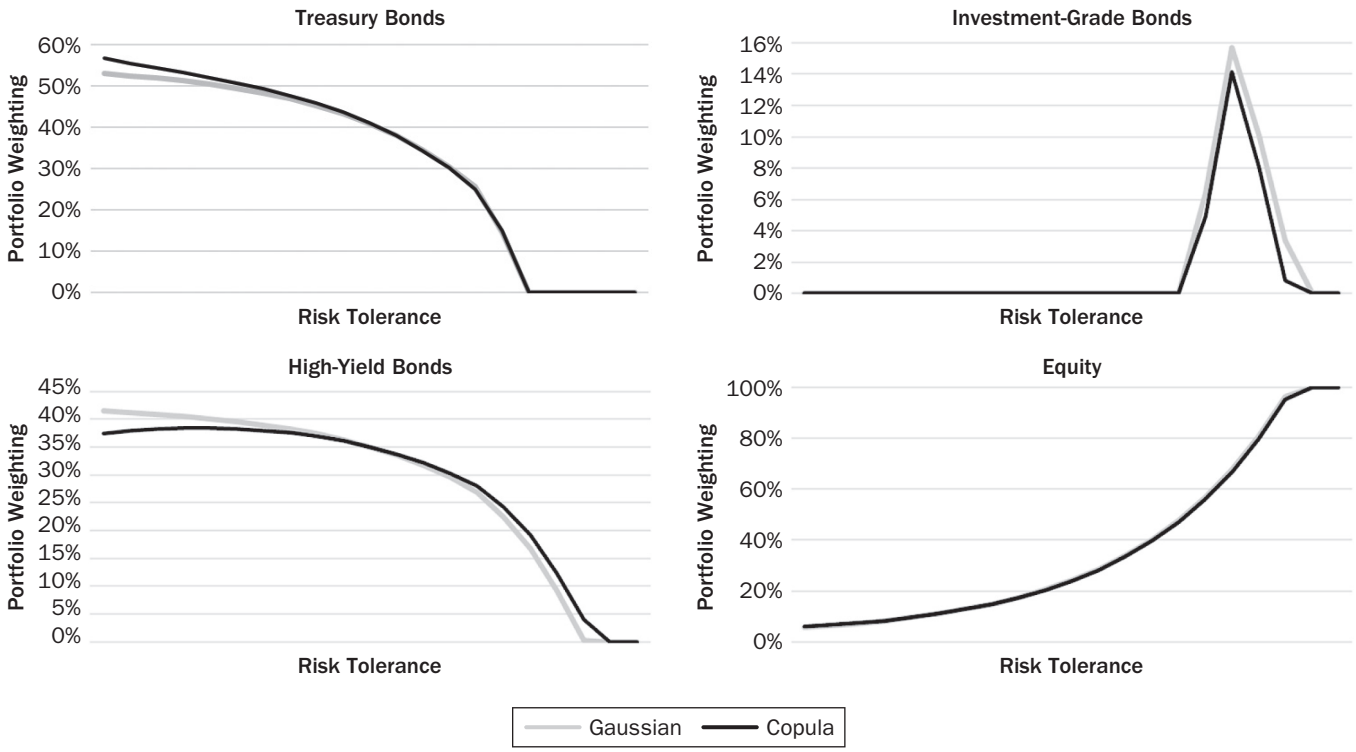
where $u(.)$ denotes the utility function. Moreover, this expression $E[u(R)|w]$ is a well-defined function of w, and so a nonlinear optimization package can readily optimize this function to find the portfolio weights that maximize expected utility of portfolio return. Because the vectors $r_1, \ldots, r_T$ have already been simulated, there is relatively little additional computation burden required to do the optimization. The great benefit of this approach is that *any* utility function can be used.

Using the 250,000 simulated returns for each asset class, we optimized the Monte Carlo approximation of the investor expected utility, using a negative exponential utility function. For comparison, we also found the optimal weights under the assumption of Gaussian returns, that is, the traditional mean–variance optimization. We found the optimal portfolio for a wide range of levels of risk aversion. The resultant portfolio weightings of the four asset classes are shown in Exhibit 7. There are several features of the data worthy of further comment: Generally, the investment behavior is similar when using both methods of portfolio construction. Note that investment-grade bonds are seldom liked by the optimizer, which we suspect is due to them being highly correlated with Treasury bonds but being less negatively correlated with equity than Treasuries.

At high levels of risk aversion, our Monte Carlo approach yields less high yield than the Markowitz portfolio. This discrepancy is due to the high skewness and kurtosis present in the high-yield asset class. As the investor's tolerance for risk increases, however, the impact of these higher moments diminishes, and the investor becomes more focused on the higher expected return of high-yield assets. Eventually,

EXHIBIT 7

Optimized Portfolio Weights



NOTES: The black lines indicate the maximum-utility portfolio weights using simulated data, and the gray lines indicate the mean–variance optimal portfolio weights. Calculations made by the authors.

with sufficient risk tolerance, the desired allocation to high yield surpasses that in the Markowitz approach. Notably, this relative increase in high yield for higher risk tolerances comes at the expense of investment-grade bonds. Interestingly, the level of equity exposure remains identical under both approaches across the entire risk tolerance spectrum. It appears that skewness has a greater influence than kurtosis in differentiating the weights in the Monte Carlo study from those in the Markowitz study.

CONCLUSION

In this article, we explain how to carry out Monte Carlo simulations in both univariate and multivariate settings and indicate how this can be of use for investment decision purposes. We offer approximate *ad hoc* methods that fit within an Excel spreadsheet, as well as precise methods that require a certain amount of computer programming. We demonstrate how MC simulation can be used to estimate nonlinear risk, particularly using downside-risk measures, and how it can be applied in a generalized portfolio optimization exercise that accounts for nonlinear asset-price behavior. MC simulation provides an elegant means to overcome the shortcomings that are inherent to the analytical specification of the portfolio optimization problem and to find solutions beyond the mean–variance optimum.

The investment cases we look at in this article are single period only. When extending to multi-period cases, intertemporal dependencies come into play, which make the simulation processes yet more complex. If dealt with methodologically,

one can introduce features of temporal asset-price behavior into an MC simulation and be in the position to analyze entire investment strategies via this method. One can analyze a buy-and-hold strategy, for instance, where the portfolio manager acts depending on certain outcomes, which thus makes the overall portfolio outcome path dependent. Strategies following the constant proportion portfolio insurance (CPPI) principle tend to be analyzed in this fashion.

Typically, in the discipline of quantitative finance, summary statistics are desired to offer single value metrics, which can give guidance to the characteristics of the return of an investment. Commonplace examples are expected return and variance or standard deviation. Also common are correlation and beta (in a CAPM sense) or other factor loadings. A severe limitation of summary statistics is that they can seldom be derived as analytical expressions, especially when applied to portfolios. Hence, the focus on linear and quadratic measures such as expected return and variance. The power of Monte Carlo simulation is that summary statistics, which offer a richer characterization of a portfolio, can be computed. Apart from the VaR, the conditional VaR and the semi-volatility discussed in this article, there are many others in which we find end-investors are interested.

As long as a summary statistic can be defined, it can be computed from a Monte Carlo simulation. When using Monte Carlo techniques, the quantitative analysts should not be constrained by the shackles of analytically derivable summary statistics but instead should be open minded to using any summary statistic that best characterizes the feature of interest in a particular situation.

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